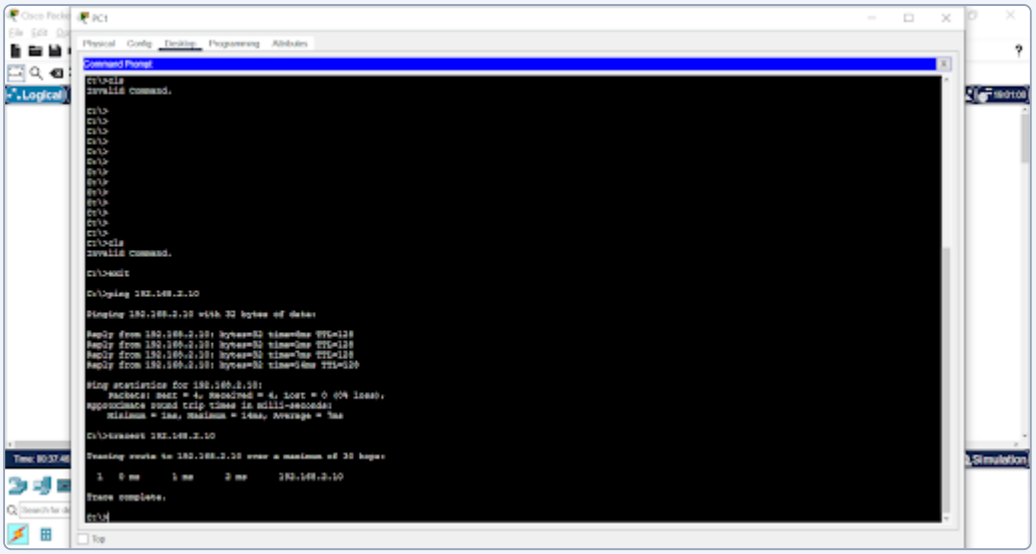
Lab 03 Report

COMPUTER & COMMUNICATION SYSTEMS ARCHITECTURE

Institution: Vilnius College of Design
Topic: Cisco Packet Tracer & Route Tracing
Date: November 2026

1. Connectivity & Route Tracing Verification



2. Lab Questions & Answers

Q1. How many hops were required to reach PC-B?

Answer: Exactly **2 Hops** were required to reach PC-B (192.168.2.10) from PC-A (192.168.1.10).

Q2. What IP is at Hop 1? Identify the device, interface and purpose.

- **IP Address:** `192.168.1.1`
- **Device:** Router-A (Router0)
- **Interface:** GigabitEthernet0/0
- **Purpose:** It serves as the Default Gateway for the local LAN subnet (`192.168.1.0/24`), receiving packets from PC-A and forwarding them outside the local network.

Q3. What IP is at Hop 2? Explain why routing uses this hop.

- **IP Address:** `10.0.0.2`
- **Device & Interface:** Router-B (Router1), Interface GigabitEthernet0/1
- **Explanation:** Routing uses this hop because Router-A has a static route configured (`ip route 192.168.2.0 255.255.255.0 10.0.0.2`). This designates `10.0.0.2` as the Next-Hop IP address through which packets must travel to reach the remote target network (`192.168.2.0/24`) where PC-B resides.